

Problem 1 (Linear Independence, Matrices) [10pts].

Consider three vectors $u, v, w \in \mathbb{R}^3$ that are **linearly independent**, and let $a, b, c, d \in \mathbb{R}^3$ be defined as follows:

$$a = u + v, \quad b = v + w, \quad c = u + w, \quad d = u - v.$$

- a. Suppose vectors u and v have unit length (i.e., $\|u\| = 1$ and $\|v\| = 1$). Determine the angle between the vectors a and d . Show your reasoning clearly to receive credit. *Hint:* Compute $a^\top d$.

Solution. The vectors a, d are orthogonal (so the angle between them is $\pi/2$) since $a^\top d = u^\top u - u^\top v + v^\top u - v^\top v = \|u\|^2 - \|v\|^2 = 0$.

- b. Again suppose vectors u and v have unit length. Give an expression for a matrix M such that $MA = I$ where $A = \begin{bmatrix} a & d \end{bmatrix} \in \mathbb{R}^{3 \times 2}$. Your expression should depend on a and d . *Hint:* use your result from **part a** about the relationship between a and d .

Solution. Since the two columns of A are orthogonal,

$$\begin{bmatrix} \frac{a^\top}{\|a\|^2} \\ \frac{d^\top}{\|d\|^2} \end{bmatrix}$$

gives a left inverse; we can check that multiplying with A gives the identity matrix. Indeed,

$$\begin{bmatrix} \frac{a^\top}{\|a\|^2} \\ \frac{d^\top}{\|d\|^2} \end{bmatrix} A = \begin{bmatrix} \frac{a^\top}{\|a\|^2} \\ \frac{d^\top}{\|d\|^2} \end{bmatrix} \begin{bmatrix} a & d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I.$$

- c. Show that nullspace of the matrix $B = \begin{bmatrix} a & b & c \end{bmatrix}$ contains only the zero vector $\mathbf{0} \in \mathbb{R}^3$. Recall that the nullspace of a matrix B is the set of all vectors x such that $Bx = 0$.

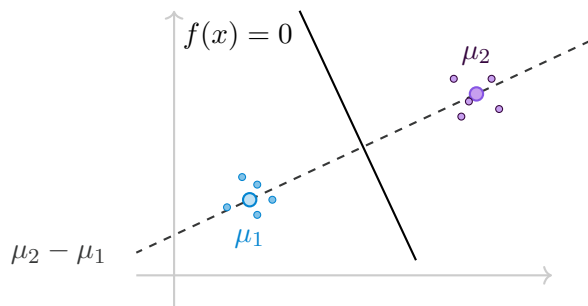
Solution. If $Bx = 0$ for some $x \in \mathbb{R}^3$, then

$$0 = x_1(u + v) + x_2(v + w) + x_3(u + w) = (x_1 + x_3)u + (x_1 + x_2)v + (x_2 + x_3)w$$

Since u, v, w are linearly independent, we have $x_1 + x_3 = 0$, $x_1 + x_2 = 0$, $x_2 + x_3 = 0$, which implies $x_1 = x_2 = x_3 = 0$, so zero is the only vector in B 's nullspace.

Problem 2 (Linear separability via cluster centroids) [10pts].

Suppose k -means with $k = 2$ yields clusters C_1, C_2 and corresponding centroids $\mu_1, \mu_2 \in \mathbb{R}^d$.



Define

$$f(x) = (\mu_1 - \mu_2)^\top x - \frac{1}{2}(\|\mu_1\|^2 - \|\mu_2\|^2).$$

- (a) Show that $f(x) = 0$ defines the perpendicular bisector of the segment between μ_1 and μ_2 . Recall from the homework (and cheatsheet) that the perpendicular bisector is the set of all points equidistant from μ_1 and μ_2 :

$$H = \{x \in \mathbb{R}^d : \|x - \mu_1\| = \|x - \mu_2\|\}.$$

Solution. Expanding $\|x - \mu_1\|^2 - \|x - \mu_2\|^2 = 0$ gives

$$\begin{aligned} 0 &= \|x - \mu_1\|^2 - \|x - \mu_2\|^2 \\ &= (x - \mu_1)^\top (x - \mu_1) - (x - \mu_2)^\top (x - \mu_2) \\ &= \|x\|^2 - 2x^\top \mu_1 + \|\mu_1\|^2 - \|x\|^2 + 2x^\top \mu_2 - \|\mu_2\|^2 \\ &= 2(\mu_2 - \mu_1)^\top x + \|\mu_1\|^2 - \|\mu_2\|^2 \\ &= -2f(x) \end{aligned}$$

so that

$$f(x) = 0 \implies x \in H = \{x \in \mathbb{R}^d : \|x - \mu_1\| = \|x - \mu_2\|\}.$$

This is the equation of a plane equidistant from both centroids.

- (b) Show that $f(x_i) \geq 0$ for all $x_i \in C_1$ and $f(x_i) \leq 0$ for all $x_i \in C_2$.

Solution. Each point in G_1 is closer to μ_1 than μ_2 , so $\|x_i - \mu_1\|^2 \leq \|x_i - \mu_2\|^2$, implying $f(x_i) \geq 0$ for all $x_i \in G_1$. The argument reverses for G_2 .

- (c) A hyperplane $\{x : n^\top x = c\}$ with normal $n \neq 0$ has signed distance

$$\text{dist}_\pm(x, H) := \frac{n^\top x - c}{\|n\|}.$$

Consider $\{x \mid f(x) = 0\}$. Use **part a** to show that

$$\text{dist}_\pm(x, H) = \frac{f(x)}{\|\mu_1 - \mu_2\|},$$

Solution. This is straightforward from part (a): set $n = \mu_1 - \mu_2$ and $c = \frac{1}{2}(\|\mu_1\|^2 - \|\mu_2\|^2)$, so

$$\boxed{\text{dist}_\pm(x, H) = \frac{f(x)}{\|\mu_1 - \mu_2\|}}.$$

- (d) Now, define the *margin*

$$\gamma = \min \left\{ \min_{i \in G_1} \frac{f(x_i)}{\|\mu_1 - \mu_2\|}, \min_{i \in G_2} \frac{-f(x_i)}{\|\mu_1 - \mu_2\|} \right\}.$$

Show that if there are no ties (every point is strictly closer to its own centroid than to the other), then $\gamma > 0$.

Solution. From part (b), $f(x_i) \geq 0$ for $i \in G_1$ and $f(x_i) \leq 0$ for $i \in G_2$. If there are no ties, then $f(x_i) > 0$ for all $i \in G_1$ and $-f(x_i) > 0$ for all $i \in G_2$, so both minima inside the definition of γ are strictly positive. Therefore

$$\boxed{\gamma > 0},$$

which gives a positive (signed) separation margin between the two clusters relative to the bisector H .

Problem 3 (Closest point to a line: orthogonal projection) [10pts].

Let $x \in \mathbb{R}^n$ and a line $L = \{a + \theta v \mid \theta \in \mathbb{R}\}$ with $v \neq 0$. Find the point z^* on the line L (i.e., $z^* \in L$) minimizing $\|x - z\|_2$.

- (a) Express z as $a + \theta v$ and write $J(\theta) = \|x - (a + \theta v)\|_2^2$ and expand $J(\theta)$ as a quadratic function/polynomial in θ .

Solution. Quadratic in θ . Let $r := x - a$. Then

$$\begin{aligned} J(\theta) &= \|x - a - \theta v\|_2^2 = \|r - \theta v\|_2^2 \\ &= (r - \theta v)^\top (r - \theta v) = \|r\|_2^2 - 2\theta v^\top r + \theta^2 \|v\|_2^2. \end{aligned}$$

Thus $J(\theta)$ is a **univariate quadratic** with leading coefficient $\|v\|_2^2 > 0$ —i.e., its a quadratic function of one variable θ .

- (b) Minimize $J(\theta)$ by taking derivative with respect to θ and solving for θ^* .

Solution. First-order condition. Differentiate:

$$J'(\theta) = -2v^\top r + 2\theta \|v\|_2^2.$$

Set $J'(\theta) = 0$:

$$\theta^* = \frac{v^\top r}{\|v\|_2^2} = \frac{v^\top (x - a)}{\|v\|_2^2}.$$

- (c) Give the closed form z^* and show that $(x - z^*)$ is orthogonal to v .

Solution. Closed-form minimizer and orthogonality. Therefore

$$z^* = a + \theta^* v = a + \frac{v^\top (x - a)}{\|v\|_2^2} v.$$

Check orthogonality:

$$x - z^* = x - \left(a + \frac{v^\top (x - a)}{\|v\|_2^2} v \right) = r - \frac{v^\top r}{\|v\|_2^2} v.$$

Then

$$v^\top (x - z^*) = v^\top r - \frac{v^\top r}{\|v\|_2^2} v^\top v = v^\top r - \frac{v^\top r}{\|v\|_2^2} \|v\|_2^2 = 0.$$

Hence $x - z^* \perp v$, i.e., the error vector is perpendicular to the line direction, as required for an orthogonal projection.

- (d) **[Extra Credit]** Use Cauchy–Schwarz—i.e., the fact that $\|a^\top b\| \leq \|a\|_2 \|b\|_2$ —to verify that your z^* indeed minimizes the distance.

Solution. Cauchy–Schwarz verification. From part (a),

$$J(\theta) = \|r\|_2^2 - 2\theta v^\top r + \theta^2 \|v\|_2^2.$$

Minimizing this quadratic gave $\theta^* = \frac{v^\top r}{\|v\|_2^2}$. To confirm that this indeed produces the *smallest possible value*, we can reason directly using the Cauchy–Schwarz inequality.

- **Method 1: Lower bound via Cauchy–Schwarz.** For any θ ,

$$\|r - \theta v\|_2^2 = \|r\|_2^2 - 2\theta v^\top r + \theta^2 \|v\|_2^2 \geq \|r\|_2^2 - 2|\theta| |v^\top r| + \theta^2 \|v\|_2^2.$$

By Cauchy–Schwarz, $|v^\top r| \leq \|v\|_2 \|r\|_2$, so for every θ ,

$$\|r - \theta v\|_2^2 \geq \|r\|_2^2 - 2|\theta| \|v\|_2 \|r\|_2 + \theta^2 \|v\|_2^2 = (\|v\|_2 |\theta| - \|r\|_2)^2 \geq 0.$$

Equality can only hold when $\theta = \frac{v^\top r}{\|v\|_2^2}$ and the inner product achieves its upper bound, i.e. when r is parallel to v .

- **Method 2: Completing the square.** Alternatively, completing the square (which encodes the same inequality) gives

$$\|r - \theta v\|_2^2 = \underbrace{\left(\|r\|_2^2 - \frac{(v^\top r)^2}{\|v\|_2^2} \right)}_{\text{constant} \geq 0 \text{ by Cauchy-Schwarz}} + \underbrace{\|v\|_2^2 \left(\theta - \frac{v^\top r}{\|v\|_2^2} \right)^2}_{\geq 0}.$$

The first term is the smallest possible value of $\|r - \theta v\|_2^2$, and it is nonnegative because Cauchy–Schwarz guarantees $(v^\top r)^2 \leq \|v\|_2^2 \|r\|_2^2$. The second term is zero exactly when $\theta = \theta^* = \frac{v^\top r}{\|v\|_2^2}$.

Since both terms are nonnegative and the second is minimized only at θ^* , it follows that

$$\|r - \theta v\|_2 \text{ is minimized at } \theta = \theta^*, \text{ with minimum value } \sqrt{\|r\|_2^2 - \frac{(v^\top r)^2}{\|v\|_2^2}}.$$

This argument uses Cauchy–Schwarz to bound how close r can lie to the line's direction v , and completing the square makes that bound exact.

Problem 4 (Linear Model with a Feature Map) [10pts].

Suppose we have the data points

$$\{(x_i, y_i)\}_{i=1}^4 = \{(-2, 4.2), (-1, 2.1), (1, 1.8), (2, 4.5)\}.$$

We want to fit a model of the form

$$y = \theta_0 + \theta_1 |x| + \theta_2 x^2,$$

where $\theta = (\theta_0, \theta_1, \theta_2)^\top$.

- (a) Write the least-squares cost function $J(\theta)$.

Solution.

$$J(\theta) = \sum_{i=1}^4 (y_i - \theta_0 - \theta_1 |x_i| - \theta_2 x_i^2)^2 = \|y - X\theta\|^2.$$

- (b) Write the problem as a least squares problem with linear model $f_\theta(x) = x^\top \theta$. In particular, identify X and y explicitly.

Solution.

$$X = \begin{bmatrix} 1 & |-2| & (-2)^2 \\ 1 & |-1| & (-1)^2 \\ 1 & |1| & 1^2 \\ 1 & |2| & 2^2 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 4 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \end{bmatrix}, \quad y = \begin{bmatrix} 4.2 \\ 2.1 \\ 1.8 \\ 4.5 \end{bmatrix}.$$

- (c) Write the normal equations and find the closed form for $\hat{\theta}$.

Solution. Normal equations:

$$X^\top X \hat{\theta} = X^\top y.$$

Solving gives $\hat{\theta} = (X^\top X)^{-1} X^\top y$, if $X^\top X$ is invertible.

- (d) Are the columns of X linearly independent for this dataset? Justify your answer. *Hint:* Linear independence is defined on the cheat sheet.

Solution. To check for linear independence, we need to check if there exists a non-trivial linear combination of the columns that equals the zero vector. Let

$$x_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \quad x_2 = \begin{bmatrix} 2 \\ 1 \\ 1 \\ 2 \end{bmatrix}, \quad x_3 = \begin{bmatrix} 4 \\ 1 \\ 1 \\ 4 \end{bmatrix}.$$

Then any linear combination of x_1, x_2, x_3 can be written as $c_1 x_1 + c_2 x_2 + c_3 x_3$ for some scalars c_1, c_2, c_3 . We have that

$$3x_2 - 2x_1 = \begin{bmatrix} 6 \\ 3 \\ 3 \\ 6 \end{bmatrix} - \begin{bmatrix} 2 \\ 2 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \\ 1 \\ 4 \end{bmatrix} = x_3.$$

Therefore, the columns of X are linearly dependent.

Problem 5 (Extra Credit Problem: Minimizing a matrix-weighted mean square distance) [5pts].

Let $x_1, \dots, x_L \in \mathbb{R}^n$, and consider a symmetric matrix $W \in \mathbb{R}^{n \times n}$ with $W \succeq 0$ (positive semidefinite). Consider the objective

$$J(z) = \sum_{i=1}^L (x_i - z)^\top W (x_i - z), \quad z \in \mathbb{R}^n.$$

- (a) Show that J is a convex quadratic in z and identify its Hessian.

Solution. Convexity and Hessian. Expand a single term using symmetry of W :

$$(x_i - z)^\top W (x_i - z) = x_i^\top W x_i - 2x_i^\top W z + z^\top W z.$$

Therefore

$$J(z) = \sum_{i=1}^L x_i^\top W x_i - 2 \left(\sum_{i=1}^L x_i \right)^\top W z + z^\top W z.$$

Hence

$$\nabla J(z) = -2LW \left(\sum_{i=1}^L x_i \right) + 2LW z = 2W \left(Lz - \sum_{i=1}^L x_i \right),$$

and

$$\nabla^2 J(z) = 2L \cdot W.$$

Since $W \succeq 0$, the Hessian is positive semidefinite and J is a convex quadratic. If $W \succ 0$, J is *strictly* convex.

- (b) Suppose $W \succ 0$ (positive definite). Show that J has a unique minimizer and find it in closed form.

Solution. Set the gradient to zero:

$$\nabla J(z^*) = 0 \iff W \left(Lz^* - \sum_{i=1}^L x_i \right) = 0.$$

Because W is invertible, this implies

$$z^* = \frac{1}{L} \sum_{i=1}^L x_i$$

(the usual weighted mean). Strict convexity (Hessian $2W \succ 0$) gives uniqueness.

- (c) Suppose only $W \succeq 0$. Characterize the (possibly non-unique) set of minimizers.

Solution. Again $\nabla J(z^*) = 0$ gives

$$W \left(Lz^* - \sum_{i=1}^L x_i \right) = 0.$$

Thus $z^* - \bar{x}_w \in \ker(W)$, where $\bar{x}_w := \frac{1}{L} \sum_i x_i$ is the weighted mean. Hence the full set of minimizers is the affine space

$$\mathcal{Z}^* = \bar{x}_w + \ker(W).$$

In particular, if W is singular, J is flat along directions $v \in \ker(W)$ (no penalty in those directions), so any $z^* = \bar{x}_w + v$ with $v \in \ker(W)$ achieves the same minimum value.

- (d) Among all minimizers in part (c), find the one with the *smallest Euclidean norm* $\|z\|_2$. Give a closed form and prove it is optimal and unique.

Solution. Let $K := \ker(W)$. Then $\mathcal{Z}^* = \bar{x}_w + K$. We want

$$\min_{z \in \mathcal{Z}^*} \|z\|_2^2 \iff \min_{v \in K} \|\bar{x}_w + v\|_2^2.$$

Decompose $\bar{x}_w$ orthogonally with respect to K :

$$\bar{x}_w = \bar{x}_K + \bar{x}_{K^\perp}, \quad \bar{x}_K \in K, \bar{x}_{K^\perp} \in K^\perp.$$

Then, for any $v \in K$,

$$\|\bar{x}_w + v\|_2^2 = \|\bar{x}_{K^\perp} + (\bar{x}_K + v)\|_2^2 = \|\bar{x}_{K^\perp}\|_2^2 + \|\bar{x}_K + v\|_2^2$$

because $K \perp K^\perp$. The term $\|\bar{x}_K + v\|_2^2$ is minimized uniquely at $v^* = -\bar{x}_K$, yielding

$$z_{\min} = \bar{x}_w + v^* = \bar{x}_{K^\perp}.$$

Thus the minimum-norm minimizer is the *orthogonal projection* of $\bar{x}_w$ onto $K^\perp$. Since $K = \ker(W)$, we have $K^\perp = \text{range}(W)$, so

$$z_{\min} = P_{\text{range}(W)} \bar{x}_w.$$

This element is unique because the quadratic in v is strictly convex.

